

1. GET Parameter ?

Django- URL- data **GET** parameter

Example:

```
javascript.snippet
```

JAVASCRIPT

```
http://localhost:8000/products/?category=electronics&sort=price
```

:

```
javascript.snippet
```

JAVASCRIPT

```
category = electronics sort = price
```

values Django- :

```
javascript.snippet
```

JAVASCRIPT

```
request.GET
```

2. request.GET.get() ?

GET parameter :

```
javascript.snippet
```

JAVASCRIPT

```
request.GET.get("parameter_name")
```

Example:

```
javascript.snippet
```

JAVASCRIPT

```
category = request.GET.get("category") sort_by = request.GET.get("sort")
```

URL :

```
javascript.snippet
```

JAVASCRIPT

```
/products/?category=electronics&sort=price
```

:

```
javascript.snippet JAVASCRIPT

category# "electronics"sort_by# "price"
```

3. Basic GET Parameter Example

views.py

```
javascript.snippet JAVASCRIPT

from django.shortcuts import renderfrom .models import Productdef product_list(request
):
    category = request.GET.get("category")    sort_by = request.GET.get("sort
")    products = Product.objects.all()    # Category filter    if category:        products = produ
product_list.html",    {        "products": products    }    )
```

4. URL Configuration

urls.py

```
javascript.snippet JAVASCRIPT

from django.urls import pathfrom . import viewsurlpatterns = [    path(
"products/"
,        views.product_list,        name="product_list"    ),]
```

GET parameter- URL pattern

:

```
javascript.snippet JAVASCRIPT

path("products/", ...)
```

URL- query parameter

Example:

```
javascript.snippet JAVASCRIPT

/products//products/?category=electronics/products/?category=electronics&sort=price
```

5. Template

product_list.html

```
javascript.snippet JAVASCRIPT

Product List{% if products %}           {% for product in products %}

{{ product.name }}                      -                {{ product.category }}

-                {{ product.price }}           {% endfor %}    {% else %}
No
products found.{% endif %}
```

6. GET Parameter URL Structure

URL- structure:

```
javascript.snippet JAVASCRIPT

/products/?category=electronics&sort=price
```

:

```
javascript.snippet JAVASCRIPT

/products/      Path?      Query parameter category=electronics      Key = categoryValue =
electronics
&      Multiple parameter separatorsort=price      Key = sortValue = price
```

:

```
javascript.snippet JAVASCRIPT

/path/?key=value&key2=value2
```

7. request.GET ?

request.GET QueryDict object

Example:

```
javascript.snippet
```

JAVASCRIPT

```
def product_list(request):    print(request.GET)
```

URL:

```
javascript.snippet
```

JAVASCRIPT

```
/products/?category=electronics&sort=price
```

conceptually:

```
javascript.snippet
```

JAVASCRIPT

```
'category': ['electronics'],    'sort': ['price']}>
```

8. request.GET.get() ?

```
javascript.snippet
```

JAVASCRIPT

```
category = request.GET.get("category")
```

parameter :

```
javascript.snippet
```

JAVASCRIPT

```
category=electronics
```

:

```
javascript.snippet
```

JAVASCRIPT

```
category == "electronics"
```

parameter :

```
javascript.snippet JAVASCRIPT

/products/
```

```
javascript.snippet JAVASCRIPT

category == None
```

```
javascript.snippet JAVASCRIPT

if category:
```

check

9. Default Value

get ()- default value

```
javascript.snippet JAVASCRIPT

category = request.GET.get("category", "all")
```

URL :

```
javascript.snippet JAVASCRIPT

/products/
```

```
javascript.snippet JAVASCRIPT

category == "all"
```

:

```
javascript.snippet JAVASCRIPT

/products/?category=electronics
:
```

```
javascript.snippet JAVASCRIPT

category == "electronics"
```

10. Multiple GET Parameters

parameter

```
javascript.snippet JAVASCRIPT

category = request.GET.get("category")sort_by = request.GET.get("sort
")search = request.GET.get("
search")
```

URL:

```
javascript.snippet JAVASCRIPT

/products/?category=electronics&sort=price&search=phone
```

Result:

```
javascript.snippet JAVASCRIPT

category# electronicsort_by# pricesearch# phone
```

11. GET Parameter Search

Example:

```
javascript.snippet JAVASCRIPT

def product_list(request):    search = request.GET.get("search
")    products = Product.objects.all()    if search:        products = products.filter(
product_list.html",        {        "products": products        }    )
```

URL:

```
javascript.snippet JAVASCRIPT

/products/?search=phone
```

product name- phone products

12. GET Parameter Category Filter

Product model-:

```
javascript.snippet JAVASCRIPT

category = models.CharField(...)
```

:

```
javascript.snippet JAVASCRIPT

category = request.GET.get("category")if category:    products = products.filter(
category=category    )
```

category ForeignKey , filter

13. ForeignKey Category Filter

:

```
javascript.snippet JAVASCRIPT

class Category(models.Model):    name = models.CharField(max_length=255)
```

:

```
javascript.snippet JAVASCRIPT

class Project(models.Model):
    category = models.ForeignKey(Category,
    on_delete
    =models.CASCADE,
    related_name="projects"
    )
    name = models.CharField(
    max_length
    =255)
```

category- ID filter :

```
javascript.snippet JAVASCRIPT

category_id = request.GET.get("category_id")
if category_id:
    projects = Project.objects
    .filter(
        category_id=category_id
    )
else:
    projects = Project.objects.all()
```

URL:

```
javascript.snippet JAVASCRIPT

/projects/?category_id=1
```

:

```
javascript.snippet JAVASCRIPT

category_id = 1
```

:

```
javascript.snippet JAVASCRIPT

Category ID = 1
```

projects

14. Complete ForeignKey GET Example

models.py

```

javascript.snippet
JAVASCRIPT

from django.db import models
from django.utils.text import slugify

class Category(models.Model):
    name = models.CharField(max_length=255, unique=True)
    slug = models.SlugField(max_length=255, unique=True, blank=True)

    def save(self, *args, **kwargs):
        if not self.slug:
            self.slug = slugify(self.name)
        super().save(*args, **kwargs)

    def __str__(self):
        return self.name

class Project(models.Model):
    category = models.ForeignKey(Category, on_delete=models.CASCADE, related_name="projects")
    name = models.CharField(max_length=255)
    description = models.TextField(blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return self.name

```

15. Project List View

views.py

```

javascript.snippet
JAVASCRIPT

from django.shortcuts import render
from .models import Project

def project_list(request):
    category_id = request.GET.get("category_id")
    if category_id:
        projects = Project.objects.filter(category_id=category_id)
    projects_html, { "projects": projects }

    return render(request, "projects.html", { "projects": projects })

```

correction

original code- :

```

javascript.snippet
JAVASCRIPT

girls = Project.objects.filter(...)

```

girls variable

naming:

```
javascript.snippet JAVASCRIPT

projects = Project.objects.filter(...)
```

code readable maintainable

16. URL

urls.py

```
javascript.snippet JAVASCRIPT

from django.urls import pathfrom . import viewsurlpatterns = [    path(
    "projects/"
    ,        views.project_list,        name="project_list"    ),]
```

17. Template

projects.html

```
javascript.snippet JAVASCRIPT

Projects    {% for project in projects %}                {{ project.name }}

-           Category: {{ project.category.name }}        {% empty %}        No
projects
found.      {% endfor %}
```

18. Result

project:

```
javascript.snippet JAVASCRIPT

http://localhost:8000/projects/
```

Category ID = 1:

```
javascript.snippet JAVASCRIPT
http://localhost:8000/projects/?category_id=1
```

Category ID = 2:

```
javascript.snippet JAVASCRIPT
http://localhost:8000/projects/?category_id=2
```

19. Tag Product Detail Page

product list product- detail page- , tag

```
javascript.snippet JAVASCRIPT
{{ product.name }} - {{ product.category }} -
{{
  product.price }}
```

- clickable

20. Product Name Clickable

```
javascript.snippet JAVASCRIPT
{{ product.name }} - {{ product.category }} - {{ product.
price
}}
```

:

```
javascript.snippet JAVASCRIPT
product.name
```

clickable

21. Detail URL Pattern

urls.py

```
javascript.snippet
```

JAVASCRIPT

```
path("product/", views.product_detail, name="product_detail")
```

:

URL integer ID

Example:

```
javascript.snippet
```

JAVASCRIPT

```
/product/5/
```

:

```
javascript.snippet
```

JAVASCRIPT

```
id = 5
```

22. Product Detail View

views.py

```
javascript.snippet
```

JAVASCRIPT

```
from django.shortcuts import render, get_object_or_404
from .models import Product

def product_detail(request, id):
    product = get_object_or_404(Product, id=id)
    return render(request, "product_detail.html", {"product": product})
```

get_object_or_404() ?

original code :

```
javascript.snippet
```

JAVASCRIPT

```
product = Product.objects.get(id=id)
```

object :

```
javascript.snippet JAVASCRIPT
Product.DoesNotExist
```

exception

get_object_or_404() object :

```
javascript.snippet JAVASCRIPT
404 Not Found
```

response

23. Product Detail Template

product_detail.html

```
javascript.snippet JAVASCRIPT
{{ product.name }}      Category: {{ product.category }}      Price: {{ product.price }}
{{ product.description }}      Back to Products
```

24. ID Detail URL

Complete flow:

```
javascript.snippet JAVASCRIPT
Product List      Click Product Name      /product/5/      product_detail(request, id
=
5)      Product.objects.get(id=5)      Product Detail Page
```

25. Slug

ID- slug URL- readable

Example:

```
javascript.snippet JAVASCRIPT

/product/5/
```

```
javascript.snippet JAVASCRIPT

/product/iphone-15/
```

26. Slug Model

```
javascript.snippet JAVASCRIPT

class Product(models.Model):
    name = models.CharField(max_length=255)
    slug = models.SlugField(unique=True)
    price = models.DecimalField(
        max_digits=10, decimal_places=2)
    def __str__(self):
        return self.name
```

27. Slug URL

urls.py

```
javascript.snippet JAVASCRIPT

path("product/", views.product_detail, name="product_detail")
```

28. Slug- Link

Template:

```
javascript.snippet JAVASCRIPT

{{ product.name }}
```

URL:

●●● javascript.snippet

JAVASCRIPT

/product/iphone-15/

29. Slug Detail View

●●● javascript.snippet

JAVASCRIPT

```
from django.shortcuts import get_object_or_404
def product_detail(request, slug):
    product = get_object_or_404(Product, slug=slug)
    return render(request, "product_detail.html", {"product": product})
```

30. ID vs Slug

ID	Slug
/product/5/	/product/iphone-15/
Short	Readable
Database ID	Meaningful URL
Simple	SEO/user-friendly
Common for internal URLs	Common for content pages

31. GET Parameter vs URL Parameter

URL Parameter

●●● javascript.snippet

JAVASCRIPT

/product/5/

```
javascript.snippet JAVASCRIPT
5
```

URL path-

Django:

```
javascript.snippet JAVASCRIPT
def product_detail(request, id):
```

GET Parameter

```
javascript.snippet JAVASCRIPT
/products/?category_id=5
```

:

```
javascript.snippet JAVASCRIPT
category_id=5
```

query string-

Django:

```
javascript.snippet JAVASCRIPT
category_id = request.GET.get("category_id")
```

```
javascript.snippet JAVASCRIPT
/product/5/          URL parameter/products/?category_id=5
parameter            GET
```

32. GET Parameter + Link

Template- category filter link :

```
javascript.snippet JAVASCRIPT
Electronics
```

Click :

```
javascript.snippet JAVASCRIPT
/projects/?category_id=1
```

View:

```
javascript.snippet JAVASCRIPT
category_id = request.GET.get("category_id")
:
```

```
javascript.snippet JAVASCRIPT
projects = Project.objects.filter(    category_id=category_id)
```

33. Dynamic Category Link

category object :

```
javascript.snippet JAVASCRIPT
{{ category.name }}
```

category dynamic URL

```
javascript.snippet JAVASCRIPT
Electronics/projects/?category_id=/projects/?category_id=/projects/?category_id=3
```

34. GET Search + Category

practical example:

```
javascript.snippet JAVASCRIPT

def project_list(request):
    category_id = request.GET.get("category_id")
    search = request.GET.get("search")
    projects = Project.objects.all()
    if category_id:
        projects = projects.filter(category_id=category_id)
    if search:
        projects = projects.filter(name__icontains=search)
    return render(request, "project_list.html", {"projects": projects})
```

URL:

```
javascript.snippet JAVASCRIPT

/projects/?category_id=1&search=django
```

```
javascript.snippet JAVASCRIPT

category_id category filtersearch project name search
```

35. GET Sorting

Example:

```
javascript.snippet JAVASCRIPT

def product_list(request):
    sort = request.GET.get("sort")
    products = Product.objects.all()
    if sort == "price":
        products = products.order_by("price")
    elif sort == "-price":
        products = products.order_by("-price")
    return render(request, "product_list.html", {"products": products})
```

URL:

```
javascript.snippet JAVASCRIPT

/products/?sort=price
```

Price ascending:

```
javascript.snippet JAVASCRIPT
/products/?sort=price
```

Price descending:

```
javascript.snippet JAVASCRIPT
/products/?sort=-price
```

Security Note

:

```
javascript.snippet JAVASCRIPT
products.order_by(sort_by)
```

sort_by user-controlled allowed fields

Better:

```
javascript.snippet JAVASCRIPT
sort = request.GET.get("sort")
allowed_sorting = { "price": "price", "-price": "-price",
                    "name": "name", "-name": "-name", }
sort_field = allowed_sorting.get(sort)
if sort_field:
    products = products.order_by(sort_field)
```

user arbitrary ordering expression

36. Final Complete Example

models.py

```
javascript.snippet JAVASCRIPT

from django.db import models
from django.utils.text import slugify

class Category(models.Model):
    name = models.CharField(max_length=255, unique=True)
    slug = models.SlugField(max_length=255, unique=True, blank=True)

    def save(self, *args, **kwargs):
        if not self.slug:
            self.slug = slugify(self.name)
        super().save(*args, **kwargs)

    def __str__(self):
        return self.name

class Project(models.Model):
    category = models.ForeignKey(Category, on_delete=models.CASCADE, related_name="projects")
    name = models.CharField(max_length=255)
    description = models.TextField(blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return self.name
```

views.py

```
javascript.snippet JAVASCRIPT

from django.shortcuts import render
from .models import Project

def project_list(request):
    category_id = request.GET.get("category_id")
    search = request.GET.get("search")
    projects = Project.objects.select_related("category")
    if category_id:
        projects = projects.filter(category_id=category_id)
    return render(request, "projects.html", {"projects": projects, "category_id": category_id, "search": search})
```

```
javascript.snippet JAVASCRIPT

select_related("category")
```

Project template-:

```
javascript.snippet JAVASCRIPT

{{ project.category.name }}
```

urls.py

```
javascript.snippet JAVASCRIPT

from django.urls import pathfrom . import viewsurlpatterns = [    path(
    "projects/"
    ,        views.project_list,        name="project_list"    ),]
```

projects.html

```
javascript.snippet JAVASCRIPT

Projects        type="text"        name="search"        placeholder="Search project
..."
    >        Search        {% for project in projects %}        {{
project
.name }}        -        Category: {{ project.category.name }}        {%
empty
%}        No projects found.        {% endfor %}
```

37. Final Summary

GET Parameter

```
javascript.snippet JAVASCRIPT

request.GET.get("name")
```

URL:

```
javascript.snippet JAVASCRIPT

/products/?name=phone
```

Multiple GET Parameters

javascript.snippet

JAVASCRIPT

```
/products/?category=electronics&sort=price
```

ForeignKey Filter

javascript.snippet

JAVASCRIPT

```
category_id = request.GET.get("category_id")
projects = Project.objects.filter(
    category_id
    =category_id)
```

Dynamic Link

javascript.snippet

JAVASCRIPT

```
{{ product.name }}
```

URL Parameter

javascript.snippet

JAVASCRIPT

```
/product/5/
```

javascript.snippet

JAVASCRIPT

```
def product_detail(request, id):
```

GET Parameter

javascript.snippet

JAVASCRIPT

```
/products/?category_id=5
```

```
javascript.snippet JAVASCRIPT

category_id = request.GET.get("category_id")
```

Slug

```
javascript.snippet JAVASCRIPT

{{ product.name }}
```

URL:

```
javascript.snippet JAVASCRIPT

/product/iphone-15/
```

Quick Memory Map

```
javascript.snippet JAVASCRIPT

Django URL Data

URL Parameter      GET Parameter      /
product
/5/                /products/?category_id=5      self.
kwargs
request.GET
"] GET.get("      kwargs["id
category_id")

Detail Page      Filter/Search/Sort
```

```
javascript.snippet JAVASCRIPT

1. request.GET.get()      GET parameter 2. self.kwargs      URL parameter 3.

{% url %}      Dynamic URL 4.      Clickable link5.
get_object_or_404
()      Safe object retrieval
```